

YEAR 7 MATHEMATICS

Chapter 4

Fractions and percentages

MIXED PRACTICE WORKSHEETS

4

How to use this booklet

- Each lesson is **one double-sided sheet**. Write your name, class and date at the top.
- Then work down the **columns**, left to right. They get harder as you go: **Easy** **Easy** **Medium** on page 1, then **Medium** **Challenging** **Challenging** on page 2.
- Most questions practise **today's skill**. A question marked **from 1.03** comes from an earlier lesson, to keep it fresh. Stuck? Look back at that lesson.
- Each question tells you what to do: read it carefully. Show your working on the lines. The first question with working in each Easy column shows you how to set it out.

WE DO

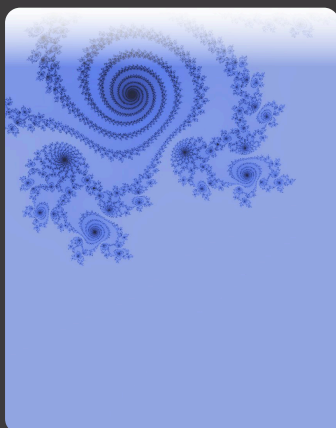
The grey box at the top. Your teacher shows the key idea with two or three examples. **Copy every step** into the space.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
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In this chapter you will

- write equivalent fractions and simplify
- order fractions
- add, subtract, multiply and divide fractions and mixed numerals
- find a fraction or percentage of a quantity
- solve fraction problems
- convert between fractions, decimals and percentages



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is a triple spiral on the edge of the top bulb. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Equivalent fractions: multiply or divide the top and the bottom by the same number.

Watch your teacher.
Copy every step.

A mixed numeral such as $2\frac{3}{4}$ is 2 wholes and $\frac{3}{4}$; as an improper fraction it is 11 quarters.

E1 Simplify $\frac{18}{24}$.

E2 Write $2\frac{3}{4}$ as an improper fraction.

E3 Write $\frac{17}{5}$ as a mixed numeral.

Easy

Questions 1–9

1 Simplify $\frac{5}{20}$.

2 Write $1\frac{1}{2}$ as an improper fraction.

3 Write $\frac{11}{8}$ as a mixed numeral.

4 Simplify $\frac{2}{10}$.

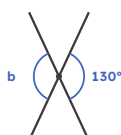
5 Write $1\frac{2}{5}$ as an improper fraction.

6 Write $\frac{7}{2}$ as a mixed numeral.

7 Simplify $\frac{5}{15}$.

8 Write $3\frac{3}{5}$ as an improper fraction.

9 Find b .



from 2.05

Easy

Questions 10–18

10 Write $3\frac{1}{2}$ as an improper fraction.

11 Write $\frac{10}{4}$ as a mixed numeral.

12 Simplify $\frac{2}{12}$.

13 $390 \div 10$

from 3.03

14 Find the missing number: $\frac{5}{6} = \frac{\square}{24}$

$\square \times \square = \square$

$\square = \square \times \square = \square$

15 Write $3\frac{1}{3}$ as an improper fraction.

16 Write $\frac{17}{5}$ as a mixed numeral.

17 List the factors of 8.

from 3.09

18 Simplify $\frac{15}{20}$.

Medium

Questions 19–26

19 Find the missing number: $\frac{5}{12} = \frac{\square}{96}$

20 Simplify $\frac{32}{40}$ fully.

21 List the first five multiples of 3.

from 3.12

22 Which of these is not equal to $\frac{2}{5}$: $\frac{5}{10}$, $\frac{6}{15}$, $\frac{4}{10}$, $\frac{8}{20}$?

23 Write $\frac{21}{5}$ as a mixed numeral.

24 $6\,996 \div 33$

from 3.05

25 Find the missing number: $\frac{5}{8} = \frac{\square}{32}$

26 Simplify $\frac{90}{117}$ fully.

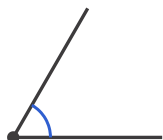
Medium

Questions 27–36

- 27 Simplify $\frac{8}{64}$ fully.

- 28 Which of these is not equal to $\frac{5}{6}$: $\frac{11}{12}$, $\frac{20}{24}$, $\frac{10}{12}$, $\frac{15}{18}$?

- 29 Classify the angle.



from 2.03

- 30 Write $\frac{19}{3}$ as a mixed numeral.

- 31 Estimate $5\,088 \div 80$.

from 3.01

- 32 Find the missing number: $\frac{2}{3} = \frac{\square}{27}$

- 33 Find the missing numbers: $\frac{4}{5} = \frac{\square}{30} = \frac{40}{\square}$

- 34 The lines are parallel. What is the co-interior angle to 142° ?

from 2.09

- 35 Simplify $\frac{8}{48}$ fully.

- 36 $XA \perp VH$. What size is the angle where they meet?

from 2.06

Challenging

Questions 37–46

- 37 Find the missing numbers: $\frac{1}{5} = \frac{\square}{20} = \frac{10}{\square}$

- 38 What fraction of an hour is 12 minutes? Simplify.

- 39 Find the digit sum of 2 311. Is 2 311 divisible by 3?

from 3.04

- 40 Chloe says $\frac{6}{10} = \frac{3}{10}$ because you halve the top. Is that right? Explain.

- 41 Write CCLXXXVIII as a number.

from 3.06

- 42 How many thirds are in $6\frac{1}{3}$?

- 43 Find the missing numbers: $\frac{1}{4} = \frac{\square}{12} = \frac{10}{\square}$

- 44 Write $2 \times 2 \times 3$ in index form.

from 3.10

- 45 What fraction of an hour is 40 minutes? Simplify.

- 46 Use the HCF to simplify the fraction $\frac{6}{10}$.

from 3.11

Challenging

Questions 47–56

- 47 What fraction of an hour is 15 minutes? Simplify.

- 48 Jack says $\frac{2}{8} = \frac{1}{8}$ because you halve the top. Is that right? Explain.

- 49 The lines are parallel. What is the alternate angle to 65° ?

from 2.08

- 50 How many quarters are in $10\frac{3}{4}$?

- 51 Draw an angle of 115° on the ray.

from 2.02



- 52 Find the missing numbers: $\frac{1}{5} = \frac{\square}{20} = \frac{7}{\square}$

- 53 Write $\frac{17}{8}$ as a mixed numeral.

- 54 The lines are parallel. An angle is 154° . What is its corresponding angle?

from 2.07

- 55 What fraction of an hour is 36 minutes? Simplify.

- 56 $\sqrt{64} + \sqrt{16}$

from 3.08

4.02 Ordering fractions

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

To compare fractions, write them with the same denominator. Then the larger numerator gives the larger fraction.

Watch your teacher.
Copy every step.

E1 Which is larger: $\frac{3}{4}$ or $\frac{5}{8}$?

E2 Order from smallest to largest: $\frac{2}{3}$, $\frac{1}{2}$, $\frac{5}{6}$

E3 Write < or >: $\frac{3}{5}$ $\square$ $\frac{2}{3}$

Easy

Questions 1–9

1 Which is larger: $\frac{3}{5}$ or $\frac{2}{5}$?

2 Which is larger: $\frac{1}{8}$ or $\frac{1}{3}$?

3 Is $\frac{7}{8}$ more or less than $\frac{1}{2}$?

4 Which is larger: $\frac{1}{7}$ or $\frac{6}{7}$?

5 Which is larger: $\frac{1}{3}$ or $\frac{1}{12}$?

6 Is $\frac{1}{7}$ more or less than $\frac{1}{2}$?

7 Which is larger: $\frac{1}{7}$ or $\frac{3}{7}$?

8 Which is larger: $\frac{1}{6}$ or $\frac{1}{2}$?

9 Write $5\frac{5}{6}$ as an improper fraction.

from 4.01

Easy

Questions 10–18

10 Which is larger: $\frac{1}{10}$ or $\frac{1}{8}$?

11 Is $\frac{8}{11}$ more or less than $\frac{1}{2}$?

12 Which is larger: $\frac{3}{4}$ or $\frac{2}{4}$?

13 How many times does 18 go into 108?

from 3.05

14 Which is larger: $\frac{1}{12}$ or $\frac{9}{10}$?

$\frac{5}{12}$ and $\frac{54}{60}$

larger: $\frac{9}{10}$

15 Which is larger: $\frac{1}{8}$ or $\frac{1}{6}$?

16 Is $\frac{1}{9}$ more or less than $\frac{1}{2}$?

17 Write $\frac{5}{4}$ as a mixed numeral.

from 4.01

18 Which is larger: $\frac{6}{9}$ or $\frac{8}{9}$?

Medium

Questions 19–26

19 Which is larger: $\frac{1}{3}$ or $\frac{1}{6}$?

20 Write < or >: $\frac{1}{3}$ $\square$ $\frac{1}{2}$

21 Write $\frac{26}{3}$ as a mixed numeral.

from 4.01

22 Order from smallest to largest: $\frac{5}{12}$, $\frac{2}{3}$, $\frac{1}{3}$

23 Which is larger: $\frac{1}{2}$ or $\frac{3}{4}$?

24 For $AB \parallel CD$, what must be true of co-interior angles?

from 2.11

25 Write < or >: $\frac{1}{2}$ $\square$ $\frac{1}{5}$

26 Order from smallest to largest: $\frac{1}{3}$, $\frac{7}{12}$, $\frac{1}{4}$

Medium

Questions 27–37

27 Write $<$ or $>$: $\frac{5}{6}$ $\square$ $\frac{3}{4}$

28 Order from smallest to largest:
 $\frac{1}{4}, \frac{2}{3}, \frac{1}{6}$

29 Write $4\frac{9}{10}$ as an improper fraction.

from 4.01

30 Which is larger: $\frac{2}{3}$ or $\frac{7}{8}$?

31 Simplify $\frac{63}{135}$ fully.

from 4.01

32 Order from largest to smallest:
 $\frac{1}{3}, 1\frac{4}{5}, 1\frac{2}{3}, 1\frac{1}{3}$

33 Write $<$ or $>$: $\frac{2}{3}$ $\square$ $\frac{5}{6}$

34 Find the LCM of 4, 6 and 5.

from 3.12

35 Order from smallest to largest:
 $\frac{11}{12}, \frac{1}{6}, \frac{5}{6}$

36 Simplify $\frac{6}{10}$.

from 4.01

37 Which is larger: $\frac{3}{4}$ or $\frac{4}{5}$?

Challenging

Questions 38–47

38 Order from largest to smallest: $1\frac{1}{2}, \frac{1}{4}, 1\frac{2}{3}, \frac{3}{4}$

39 Write a fraction between $\frac{3}{4}$ and $\frac{5}{6}$.

40 Write $\frac{8}{3}$ as a mixed numeral.

from 4.01

41 Which is closer to 1: $\frac{7}{8}$ or $\frac{5}{6}$?

42 The lines are parallel. What is the co-interior angle to 57° ?

from 2.09

43 Order from largest to smallest: $\frac{2}{3}, 1\frac{2}{3}, 1\frac{1}{2}, \frac{7}{10}$

44 Write a fraction between $\frac{1}{3}$ and $\frac{1}{2}$.

45 Write $\frac{11}{8}$ as a mixed numeral.

from 4.01

46 Which is closer to 1: $\frac{9}{10}$ or $\frac{11}{12}$?

47 Simplify $\frac{12}{84}$ fully.

from 4.01

Challenging

Questions 48–57

48 Write a fraction between $\frac{1}{4}$ and $\frac{1}{3}$.

49 Which is closer to 1: $\frac{9}{10}$ or $\frac{4}{5}$?

50 $CE \perp UZ$. What size is the angle where they meet?

from 2.06

51 Order from largest to smallest: $1\frac{2}{5}, \frac{3}{4}, \frac{1}{4}, \frac{1}{3}$

52 Write $4\frac{1}{8}$ as an improper fraction.

from 4.01

53 Order from smallest to largest: $\frac{11}{12}, \frac{1}{4}, \frac{1}{6}$

54 Write a fraction between $\frac{2}{5}$ and $\frac{1}{2}$.

55 Write $\frac{32}{9}$ as a mixed numeral.

from 4.01

56 Which is closer to 1: $\frac{4}{5}$ or $\frac{11}{12}$?

57 Find the supplement of 59° .

from 2.04

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

To add or subtract fractions, make the denominators the same. Then add or subtract the numerators and keep the denominator. Simplify the answer.

Watch your teacher.
Copy every step.

E1 $\frac{3}{8} + \frac{1}{8}$

E2 $\frac{2}{3} + \frac{1}{4}$

E3 $\frac{5}{6} - \frac{1}{4}$

Easy

Questions 1–16

1 $\frac{2}{8} + \frac{1}{8}$

2 $\frac{5}{6} - \frac{3}{6}$

3 $1 - \frac{2}{6}$

4 $\frac{7}{11} + \frac{3}{11}$

5 $\frac{6}{11} - \frac{3}{11}$

6 $1 - \frac{6}{7}$

7 $\frac{2}{8} + \frac{5}{8}$

8 $\frac{5}{9} - \frac{3}{9}$

9 Which is larger: $\frac{1}{9}$ or $\frac{1}{11}$?

from 4.02

10 $1 - \frac{5}{11}$

11 $\frac{5}{11} + \frac{3}{11}$

12 $\frac{4}{5} - \frac{1}{5}$

13 $1 - \frac{4}{6}$

14 $\frac{1}{12} + \frac{3}{12}$

15 $\frac{3}{9} - \frac{1}{9}$

16 $1 - \frac{8}{11}$

Easy

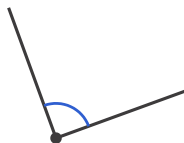
Questions 17–27

17 $\frac{6}{8} - \frac{1}{8}$

18 $1 - \frac{6}{9}$

19 $\frac{2}{5} + \frac{3}{5}$

20 Classify the angle.



from 2.03

21 $\frac{4}{5} + \frac{14}{15}$
= $\frac{12}{15} + \frac{14}{15}$
= $\frac{26}{15}$

22 $\frac{5}{10} - \frac{4}{10}$

23 $1 - \frac{4}{12}$

24 Write $3\frac{3}{4}$ as an improper fraction.

from 4.01

25 $\frac{6}{12} + \frac{3}{12}$

26 $\frac{4}{5} - \frac{7}{15}$

27 $\frac{2}{3} - \frac{1}{3}$

Medium

Questions 28–34

28 $\frac{1}{3} + \frac{8}{9}$

29 $\frac{5}{6} + \frac{3}{4}$

30 Order from smallest to largest: $\frac{1}{2}$, $\frac{1}{3}$, $\frac{3}{4}$

from 4.02

31 Jack ate $\frac{1}{3}$ of a pizza and Jo ate $\frac{1}{4}$. What fraction did they eat altogether?

32 $\frac{3}{5} - \frac{7}{15}$

33 Find the digit sum of 5 840. Is 5 840 divisible by 3?

from 3.04

34 $\frac{1}{4} + \frac{7}{10}$

4.03 Adding and subtracting fractions (continued)

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Medium

Questions 35–43

35 $\frac{1}{2} + \frac{4}{5}$

36 Priya ate $\frac{1}{3}$ of a pizza and Sam ate $\frac{1}{4}$. What fraction did they eat altogether?

37 Write $\frac{14}{4}$ as a mixed numeral.

from 4.01

38 $\frac{3}{5} - \frac{1}{15}$

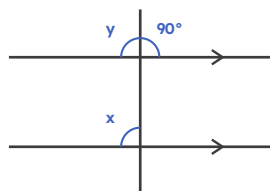
39 Which is larger: $\frac{1}{10}$ or $\frac{1}{5}$?

from 4.02

40 $\frac{1}{4} + \frac{1}{3} - \frac{1}{12}$

41 $\frac{1}{6} + \frac{1}{10}$

42 Find y , then x . Give reasons.



from 2.10

43 Priya ate $\frac{1}{2}$ of a pizza and Bo ate $\frac{1}{6}$. What fraction did they eat altogether?

Challenging

Questions 44–53

44 $\frac{1}{3} + \frac{1}{4} - \frac{1}{6}$

45 $\frac{2}{3} + \frac{5}{6}$. Write the answer as a mixed numeral.

46 Write $<$ or $>$: $\frac{5}{8} \square \frac{9}{10}$

from 4.02

47 How much more is $\frac{1}{5}$ than $\frac{1}{6}$?

48 How many times does 25 go into 75?

from 3.05

49 Find the missing fraction: $\frac{3}{4} + \square = \frac{7}{8}$

50 $\frac{1}{3} + \frac{1}{4} - \frac{1}{12}$

51 Write $1\frac{1}{12}$ as an improper fraction.

from 4.01

52 $\frac{2}{3} + \frac{5}{8}$. Write the answer as a mixed numeral.

53 Order from smallest to largest: $\frac{1}{4}, \frac{11}{12}, \frac{1}{12}$

from 4.02

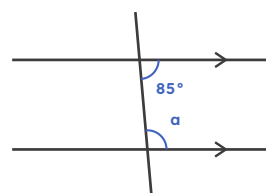
Challenging

Questions 54–61

54 $\frac{2}{3} + \frac{3}{8}$. Write the answer as a mixed numeral.

55 How much more is $\frac{5}{6}$ than $\frac{5}{8}$?

56 Find a . Give a reason.



from 2.09

57 Find the missing fraction: $\frac{2}{3} + \square = \frac{7}{8}$

58 Write $\frac{10}{3}$ as a mixed numeral.

from 4.01

59 $\frac{1}{4} + \frac{1}{6} - \frac{1}{12}$

60 Lena ate $\frac{1}{3}$ of a pizza and Jo ate $\frac{1}{5}$. What fraction did they eat altogether?

61 Which is larger: $\frac{1}{3}$ or $\frac{5}{12}$?

from 4.02

4.04 Adding and subtracting mixed numerals

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PAGE 1 OF 2

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

Add the whole numbers, then add the fractions. Or change each mixed numeral to an improper fraction first. When subtracting, if the fraction part is too small, change to improper fractions.

E1 $2\frac{1}{5} + 1\frac{3}{5}$

E2 $3\frac{1}{4} - 1\frac{3}{4}$

E3 $1\frac{2}{3} + 2\frac{1}{2}$

Easy

Questions 1–16

1 $1\frac{3}{7} + 2\frac{1}{7}$

2 $5\frac{6}{7} - 1\frac{5}{7}$

3 $5 - \frac{2}{7}$

4 $4\frac{1}{5} + 2\frac{2}{5}$

5 $3\frac{8}{9} - 1\frac{7}{9}$

6 $4 - \frac{2}{7}$

7 $4\frac{1}{4} + 4\frac{1}{4}$

8 $5\frac{8}{9} - 1\frac{7}{9}$

9 $\frac{2}{3} - \frac{1}{3}$

from 4.03

10 $5 - \frac{1}{4}$

11 $5\frac{5}{9} + 1\frac{2}{9}$

12 $5\frac{4}{5} - 2\frac{3}{5}$

13 $3 - \frac{2}{5}$

14 $3\frac{1}{9} + 2\frac{5}{9}$

15 $5\frac{4}{9} - 3\frac{2}{9}$

16 $6 - \frac{2}{5}$

Easy

Questions 17–27

17 $8\frac{4}{5} - 4\frac{3}{5}$

18 $5 - \frac{4}{7}$

19 $7\frac{1}{9} + 1\frac{1}{9}$

20 To draw a reflex angle of 270° , what angle do you draw first?

from 2.02

21 $4\frac{1}{8} - 2\frac{3}{8}$

$= \frac{33}{8} - \frac{19}{8}$

$= \frac{14}{8} = \frac{7}{4} = 1\frac{3}{4}$

22 $8\frac{7}{9} - 7\frac{5}{9}$

23 $3 - \frac{1}{3}$

24 Which is larger: $\frac{1}{5}$ or $\frac{1}{10}$?

from 4.02

25 $1\frac{2}{5} + 1\frac{1}{5}$

26 $3\frac{5}{8} - 2\frac{7}{8}$

27 $3\frac{4}{5} - 1\frac{2}{5}$

Medium

Questions 28–34

28 $3\frac{2}{5} - 2\frac{3}{5}$

29 $2\frac{2}{3} + 1\frac{2}{5}$

30 Simplify $\frac{24}{56}$ fully.

from 4.01

31 A pipe is $6\frac{1}{2}$ m long. $2\frac{1}{2}$ m is cut off. How much is left?

32 $6\frac{1}{6} - 5\frac{5}{6}$

33 Write 23 in Roman numerals.

from 3.06

34 $2\frac{1}{2} + 2\frac{2}{5}$

Medium

Questions 35–43

35 $2\frac{1}{2} + 3\frac{2}{5}$

36 A pipe is $4\frac{1}{2}$ m long. $1\frac{1}{2}$ m is cut off. How much is left?

37 $1 - \frac{1}{11}$

from 4.03

38 $6\frac{5}{8} - 4\frac{7}{8}$

39 Order from smallest to largest:
 $\frac{1}{6}, \frac{7}{12}, \frac{1}{3}$

from 4.02

40 $3\frac{3}{4} - 2\frac{1}{2}$

41 $5\frac{2}{3} + 4\frac{5}{6}$

42 The highest temperature was 12°C . The lowest was -15°C . What is the difference?

from 1.10

43 A pipe is $5\frac{1}{2}$ m long. $2\frac{1}{2}$ m is cut off. How much is left?

Challenging

Questions 44–52

44 $5\frac{3}{4} - 4\frac{1}{6}$

45 A rectangle is $7\frac{1}{4}$ m long and $1\frac{3}{4}$ m wide. Find its perimeter.

46 $\frac{2}{4} + \frac{5}{12}$

from 4.03

47 $3\frac{1}{4} + 3\frac{5}{8} + 1$

48 Find the digit sum of 4 942. Is 4 942 divisible by 3?

from 3.04

49 $4\frac{1}{3} - 2\frac{1}{2}$

50 A rectangle is $7\frac{1}{2}$ m long and $1\frac{1}{2}$ m wide. Find its perimeter.

51 Which is larger: $\frac{2}{12}$ or $\frac{7}{12}$?

from 4.02

52 $3\frac{1}{3} + 1\frac{3}{4} + 2$

Challenging

Questions 53–61

53 A rectangle is $6\frac{1}{2}$ m long and $2\frac{1}{2}$ m wide. Find its perimeter.

54 $3\frac{1}{2} + 3\frac{3}{8} + 2$

55 $\sqrt{9} + \sqrt{16}$

from 3.08

56 $4\frac{1}{3} - 1\frac{3}{4}$

57 $\frac{3}{4} - \frac{1}{4}$

from 4.03

58 A pipe is $5\frac{1}{2}$ m long. $1\frac{1}{2}$ m is cut off. How much is left?

59 A rectangle is $6\frac{3}{4}$ m long and $4\frac{3}{4}$ m wide. Find its perimeter.

60 Write < or >: $\frac{7}{10} \square \frac{1}{2}$

from 4.02

61 $2\frac{3}{4} + 1\frac{1}{6} + 2$

4.05 Fraction of a quantity

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

To find a fraction of an amount, divide by the denominator, then multiply by the numerator: $\frac{3}{4}$ of 20 = $20 \div 4 \times 3 = 15$. Change to a smaller unit first when you need to.

Watch your teacher.
Copy every step.

E1 $\frac{1}{5}$ of 35

E2 $\frac{3}{4}$ of 20

E3 $\frac{2}{3}$ of an hour, in minutes

Easy

Questions 1–13

1 $\frac{1}{3}$ of 15

2 $\frac{2}{3}$ of 6

3 What is a fifth of 15?

4 $\frac{1}{12}$ of 132

5 $\frac{3}{5}$ of 45

6 What is a third of 45?

7 $\frac{1}{10}$ of 80

8 $\frac{3}{7}$ of 35

9 Write $1\frac{3}{4}$ as an improper fraction.

from 4.01

10 What is a third of 6?

11 $\frac{1}{10}$ of 100

12 $\frac{5}{6}$ of 18

13 What is a fifth of 55?

Easy

Questions 14–22

14 $\frac{4}{5}$ of 20

15 What is a fifth of 70?

16 $\frac{1}{12}$ of 144

17 Are the lines in train tracks parallel, perpendicular or neither?

from 2.06

18 $\frac{2}{3}$ of 18

$\frac{\quad}{\quad} \div \frac{\quad}{\quad} = \frac{\quad}{\quad}$

$\frac{\quad}{\quad} \times \frac{\quad}{\quad} = \frac{\quad}{\quad}$

19 $\frac{5}{6}$ of 12

20 What is a fifth of 50?

21 $3\frac{4}{5} - 2\frac{3}{5}$

from 4.04

22 $\frac{1}{6}$ of 24

Medium

Questions 23–30

23 $\frac{7}{8}$ of 48

24 $\frac{3}{10}$ of 1 kg, in grams

25 $\frac{2}{3} - \frac{2}{5}$

from 4.03

26 $\frac{3}{4}$ of 24 students are girls. How many is that?

27 $\frac{2}{3}$ of 30

28 List the factors of 15.

from 3.09

29 $\frac{2}{3}$ of a day, in hours

30 $\frac{3}{5}$ of 25 players are under 12. How many is that?

Medium

Questions 31–41

31 $\frac{1}{2}$ of \$1, in cents

32 $\frac{4}{5}$ of 30 eggs are brown. How many is that?

33 Which is larger: $\frac{1}{4}$ or $\frac{1}{3}$?

from 4.02

34 $\frac{9}{10}$ of 240

35 Which of these is not equal to $\frac{3}{4}$: $\frac{6}{8}$, $\frac{12}{16}$, $\frac{7}{8}$, $\frac{9}{12}$?

from 4.01

36 $\frac{1}{7}$ of a number is 15. What is the number?

37 $\frac{1}{2}$ of an hour, in minutes

38 $1\,494 \div 6$

from 3.03

39 $\frac{1}{6}$ of 24 players are under 12. How many is that?

40 $2 - \frac{5}{6}$

from 4.04

41 $\frac{2}{3}$ of 57

Challenging

Questions 42–50

42 $\frac{1}{3}$ of a number is 6. What is the number?

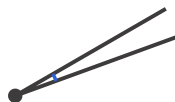
43 $\frac{3}{5}$ of a number is 9. What is the number?

44 Aisha ate $\frac{1}{3}$ of a pizza and Jo ate $\frac{1}{5}$. What fraction did they eat altogether?

from 4.03

45 Which is more: $\frac{2}{3}$ of 60 or $\frac{3}{4}$ of 36?

46 Classify the angle.



from 2.03

47 A \$12 jacket is $\frac{1}{3}$ off. What is the sale price?

48 Ruby has \$60. $\frac{1}{3}$ is spent on food and $\frac{1}{4}$ on transport. How much is left?

49 Is $\frac{1}{5}$ more or less than $\frac{1}{2}$?

from 4.02

50 $\frac{1}{4}$ of a number is 14. What is the number?

Challenging

Questions 51–60

51 $\frac{2}{3}$ of a number is 12. What is the number?

52 Which is more: $\frac{2}{3}$ of 36 or $\frac{3}{4}$ of 72?

53 Write CDXXX as a number.

from 3.06

54 A \$72 jacket is $\frac{1}{3}$ off. What is the sale price?

55 $3\frac{1}{5} + 2\frac{3}{5}$

from 4.04

56 Ali has \$24. $\frac{1}{3}$ is spent on food and $\frac{1}{4}$ on transport. How much is left?

57 $\frac{1}{5}$ of a number is 10. What is the number?

58 $\frac{3}{5} + \frac{1}{15}$

from 4.03

59 $\frac{1}{5}$ of 20 students are girls. How many is that?

60 The lines are parallel. What is the co-interior angle to 106° ?

from 2.09

4.06 Multiplying fractions

YEAR 7 · CHAPTER 4: FRACTIONS AND PERCENTAGES · PAGE 1 OF 2

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

To multiply fractions, multiply the numerators and multiply the denominators. Cancel first to keep the numbers small. Change mixed numerals to improper fractions first.

Watch your teacher.
Copy every step.

E1 $\frac{2}{3} \times \frac{4}{5}$

E2 $\frac{3}{4} \times \frac{8}{9}$

E3 $1\frac{1}{2} \times 2\frac{2}{3}$

Easy

Questions 1–16

1 $\frac{1}{9} \times \frac{1}{5}$

2 $9 \times \frac{3}{7}$

3 $\frac{1}{4}$ of $\frac{1}{4}$

4 $\frac{4}{7} \times \frac{2}{5}$

5 $6 \times \frac{2}{3}$

6 $\frac{1}{2}$ of $\frac{1}{4}$

7 $\frac{2}{3} \times \frac{7}{8}$

8 $3 \times \frac{1}{3}$

9 $\frac{3}{5}$ of 25

from 4.05

10 $\frac{1}{3}$ of $\frac{1}{5}$

11 $\frac{1}{3} \times \frac{6}{7}$

12 $7 \times \frac{4}{7}$

13 $\frac{1}{4}$ of $\frac{1}{3}$

14 $\frac{1}{9} \times \frac{2}{9}$

15 $3 \times \frac{2}{5}$

16 $\frac{1}{2}$ of $\frac{1}{6}$

Easy

Questions 17–27

17 $9 \times \frac{1}{3}$

18 $\frac{1}{2}$ of $\frac{1}{2}$

19 $\frac{2}{3} \times \frac{1}{3}$

20 Evaluate 2^2 .

from 3.07

21 $\frac{3}{7} \times \frac{4}{21}$

cancel ____: $\frac{1}{\underline{\quad}} \times \frac{4}{\underline{\quad}}$

= $\frac{4}{\underline{\quad}}$

22 $9 \times \frac{1}{5}$

23 $\frac{1}{3}$ of $\frac{1}{6}$

24 $8\frac{4}{5} - 5\frac{1}{5}$

from 4.04

25 $\frac{3}{8} \times \frac{1}{6}$

26 $\frac{3}{7} \times \frac{1}{9}$

27 $7 \times \frac{6}{7}$

Medium

Questions 28–35

28 $\frac{6}{7} \times \frac{3}{10}$

29 $3\frac{1}{3} \times \frac{3}{5}$

30 Simplify $\frac{16}{24}$ fully.

from 4.01

31 $\frac{1}{4} \times 16$

32 $\frac{5}{7} \times \frac{1}{30}$

33 $\frac{4}{5} + \frac{2}{3}$

from 4.03

34 $3\frac{2}{5} \times \frac{1}{3}$

35 $\frac{1}{6} \times 12$

Medium

Questions 36–46

36 $2\frac{1}{2} \times \frac{1}{2}$

37 $\frac{1}{7} \times 35$

38 The lines are parallel. What is the alternate angle to 103° ?

from 2.08

39 $\frac{2}{5} \times \frac{1}{8}$

40 Which is larger: $\frac{1}{8}$ or $\frac{1}{7}$?

from 4.02

41 $1\frac{1}{2} \times 1\frac{2}{3}$

42 $3\frac{1}{4} \times \frac{2}{5}$

43 What is a fifth of 70?

from 4.05

44 $\frac{1}{3} \times 6$

45 A pipe is $5\frac{1}{2}$ m long. $2\frac{1}{2}$ m is cut off. How much is left?

from 4.04

46 $\frac{4}{7} \times \frac{1}{24}$

Challenging

Questions 47–56

47 $3\frac{1}{3} \times 3\frac{2}{5}$

48 Find the area of a rectangle $3\frac{1}{2}$ m long and $1\frac{1}{2}$ m wide.

49 $\sqrt{64} + \sqrt{36}$

from 3.08

50 Is $\frac{3}{5} \times \frac{1}{2}$ larger or smaller than $\frac{3}{5}$? Explain.

51 $1 - \frac{2}{8}$

from 4.03

52 $3\frac{3}{5} \times 3\frac{1}{5}$

53 Find the area of a rectangle $4\frac{1}{2}$ m long and $3\frac{1}{2}$ m wide.

54 Which of these is not equal to $\frac{2}{3}$: $\frac{4}{6}$, $\frac{6}{9}$, $\frac{8}{12}$, $\frac{5}{6}$?

from 4.01

55 Is $\frac{4}{5} \times \frac{5}{6}$ larger or smaller than $\frac{4}{5}$? Explain.

56 $\frac{2}{5}$ of 100

from 4.05

Challenging

Questions 57–66

57 Find the area of a rectangle $5\frac{1}{3}$ m long and $1\frac{2}{3}$ m wide.

58 Is $\frac{2}{3} \times \frac{3}{4}$ larger or smaller than $\frac{2}{3}$? Explain.

59 Find the supplement of 81° .

from 2.04

60 $1\frac{1}{2} \times 1\frac{3}{4}$

61 Is $\frac{4}{5}$ more or less than $\frac{1}{2}$?

from 4.02

62 $\frac{3}{5} \times 10$

63 Find the area of a rectangle $4\frac{1}{3}$ m long and $3\frac{2}{3}$ m wide.

64 $3\frac{1}{7} + 1\frac{2}{7}$

from 4.04

65 $3\frac{2}{3} \times 2\frac{1}{4}$

66 $\frac{4}{5} + \frac{13}{15}$

from 4.03

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

To divide by a fraction, multiply by its reciprocal (turn it upside down): keep, change, flip. Change mixed numerals to improper fractions first.

Watch your teacher.
Copy every step.

E1 Write the reciprocal of $\frac{3}{7}$

E2 $\frac{2}{3} \div \frac{4}{5}$

E3 $2\frac{1}{2} \div 1\frac{1}{4}$

Easy

Questions 1–9

1 Write the reciprocal of $\frac{11}{6}$.

2 How many $\frac{1}{5}$ s are in 6? Work out $6 \div \frac{1}{5}$.

3 $\frac{1}{7} \div 5$

4 Write the reciprocal of $\frac{3}{4}$.

5 How many $\frac{1}{4}$ s are in 2? Work out $2 \div \frac{1}{4}$.

6 $\frac{4}{7} \div 3$

7 Write the reciprocal of $\frac{2}{9}$.

8 How many $\frac{1}{2}$ s are in 5? Work out $5 \div \frac{1}{2}$.

9 $3 \times \frac{1}{4}$

from 4.06

Easy

Questions 10–18

10 How many $\frac{1}{4}$ s are in 3? Work out $3 \div \frac{1}{4}$.

11 $\frac{7}{8} \div 4$

12 Write the reciprocal of $\frac{6}{7}$.

13 Find the supplement of 39° .

from 2.04

14 $\frac{3}{7} \div \frac{1}{4}$

$$= \frac{3}{\quad} \times \frac{4}{\quad}$$

$$= \frac{\quad}{\quad}$$

15 How many $\frac{1}{3}$ s are in 4? Work out $4 \div \frac{1}{3}$.

16 $\frac{8}{9} \div 3$

17 $\frac{4}{7}$ of 35

from 4.05

18 Write the reciprocal of $\frac{5}{12}$.

Medium

Questions 19–25

19 $\frac{1}{7} \div \frac{6}{7}$

20 How many $\frac{1}{2}$ -cup scoops are in 4 cups of flour?

21 Simplify $\frac{8}{44}$ fully.

from 4.01

22 $5 \div \frac{2}{3}$

23 $\frac{1}{8} \div \frac{1}{8}$

24 $3\frac{3}{4} + 3\frac{5}{6}$

from 4.04

25 How many $\frac{1}{3}$ -cup scoops are in 2 cups of flour?

Medium

Questions 26–35

- 26 How many $\frac{1}{2}$ -cup scoops are in 3 cups of flour?

27 $6 \div \frac{4}{5}$

- 28 Find b .



from 2.05

29 $\frac{1}{3} \div \frac{4}{7}$

- 30 Which is larger: $\frac{1}{6}$ or $\frac{1}{10}$?

from 4.02

31 $3\frac{1}{4} \div 1\frac{2}{3}$

- 32 How many $\frac{1}{2}$ -cup scoops are in 5 cups of flour?

33 $\frac{1}{4}$ of $\frac{1}{5}$

from 4.06

34 $5 \div \frac{5}{6}$

- 35 $\frac{2}{5}$ of 35 eggs are brown. How many is that?

from 4.05

Challenging

Questions 36–44

36 $1\frac{1}{4} \div 2\frac{3}{4}$

- 37 A ribbon 6 m long is cut into pieces $\frac{3}{4}$ m long. How many pieces?

38 Evaluate 5^4 .

from 3.07

- 39 Is $\frac{2}{3} \div \frac{1}{3}$ larger or smaller than $\frac{2}{3}$? Explain.

40 $3 - \frac{1}{2}$

from 4.04

41 $2\frac{1}{2} \div 2\frac{2}{3}$

- 42 A ribbon $7\frac{1}{2}$ m long is cut into pieces $\frac{3}{4}$ m long. How many pieces?

43 $\frac{1}{2} - \frac{1}{3}$

from 4.03

- 44 Is $\frac{3}{4} \div \frac{1}{2}$ larger or smaller than $\frac{3}{4}$? Explain.

Challenging

Questions 45–54

- 45 A ribbon $4\frac{1}{2}$ m long is cut into pieces $\frac{3}{8}$ m long. How many pieces?

- 46 Is $\frac{5}{6} \div \frac{1}{2}$ larger or smaller than $\frac{5}{6}$? Explain.

47 $91 \times 1\,000$

from 3.02

48 $2\frac{1}{2} \div 1\frac{3}{5}$

- 49 Write $\frac{18}{5}$ as a mixed numeral.

from 4.01

50 $2 \div \frac{6}{7}$

- 51 A ribbon 10 m long is cut into pieces $\frac{2}{3}$ m long. How many pieces?

52 $\frac{1}{2}$ of 6

from 4.05

53 $2\frac{2}{5} \div 2\frac{4}{5}$

54 $4\frac{1}{5} - 2\frac{4}{5}$

from 4.04

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Read the problem and decide: add, subtract, multiply or divide? Write the calculation, then answer in a sentence. The whole is 1.

Watch your teacher.
Copy every step.

E1 Sam had \$60 and spent $\frac{2}{5}$ of it. How much is left?

E2 In a class, $\frac{1}{2}$ walk and $\frac{1}{3}$ ride. The rest take the bus. What fraction take the bus?

E3 How many $\frac{3}{4}$ m pieces can be cut from 6 m?

Easy

Questions 1–7

1 $\frac{4}{5}$ of a bottle of water is drunk. What fraction is left?

2 A class has 10 girls and 10 boys. What fraction of the class are girls?

$$\frac{10}{\quad} = \frac{1}{\quad}$$

3 A 21 cm ribbon is cut into 3 equal pieces. What fraction of the ribbon is each piece, and how long is it?

4 $\frac{1}{3}$ of a book is read. What fraction is left?

5 A class has 8 girls and 9 boys. What fraction of the class are girls?

6 An 18 cm ribbon is cut into 2 equal pieces. What fraction of the ribbon is each piece, and how long is it?

7 $\frac{4}{5}$ of a tank of petrol is used. What fraction is left?

Easy

Questions 8–14

8 A class has 9 girls and 14 boys. What fraction of the class are girls?

$$\frac{9}{\quad} = \frac{9}{\quad}$$

9 A 27 cm ribbon is cut into 3 equal pieces. What fraction of the ribbon is each piece, and how long is it?

10 $\frac{2}{3}$ of a tank of petrol is used. What fraction is left?

11 For $AB \parallel CD$, what must be true of corresponding angles?

from 2.11

12 Ruby had \$36 and spent $\frac{1}{3}$ of it. How much is left?

13 A class has 15 girls and 14 boys. What fraction of the class are girls?

14 A 9 cm ribbon is cut into 3 equal pieces. What fraction of the ribbon is each piece, and how long is it?

Medium

Questions 15–20

15 Kai had \$20 and spent $\frac{1}{4}$ of it. How much is left?

16 40 students: $\frac{1}{2}$ walk to school and $\frac{1}{5}$ ride. The rest take the bus. How many take the bus?

17 Simplify $\frac{66}{72}$ fully.

from 4.01

18 A garden is $\frac{1}{4}$ lawn and $\frac{1}{6}$ vegetables. The rest is flowers. What fraction is flowers?

19 Zac had \$20 and spent $\frac{3}{4}$ of it. How much is left?

20 $1\frac{4}{5} \times \frac{1}{2}$

from 4.06

Medium

Questions 21–29

- 21 80 students: $\frac{1}{4}$ walk to school and $\frac{1}{5}$ ride. The rest take the bus. How many take the bus?

- 22 A garden is $\frac{1}{2}$ lawn and $\frac{1}{8}$ vegetables. The rest is flowers. What fraction is flowers?

- 23 Find the supplement of 8° .

from 2.04

- 24 Mia had \$33 and spent $\frac{1}{3}$ of it. How much is left?

- 25 Which is larger: $\frac{1}{3}$ or $\frac{1}{5}$?

from 4.02

- 26 Noah had \$48. $\frac{1}{3}$ was spent on Monday and $\frac{1}{4}$ of what was left on Tuesday. How much is left?

- 27 100 students: $\frac{1}{4}$ walk to school and $\frac{1}{5}$ ride. The rest take the bus. How many take the bus?

- 28 What is half of 4?

from 4.05

- 29 A garden is $\frac{1}{2}$ lawn and $\frac{1}{3}$ vegetables. The rest is flowers. What fraction is flowers?

Challenging

Questions 30–37

- 30 Aisha had \$120. $\frac{1}{3}$ was spent on Monday and $\frac{1}{4}$ of what was left on Tuesday. How much is left?

- 31 After spending $\frac{3}{5}$ of the money, \$120 was left. How much was there at the start?

- 32 Evaluate 5^4 .

from 3.07

- 33 How many $\frac{3}{4}$ m lengths can be cut from a 9 m pipe?

- 34 $\frac{1}{6}$ of $\frac{1}{5}$

from 4.06

- 35 Kai had \$84. $\frac{1}{3}$ was spent on Monday and $\frac{1}{4}$ of what was left on Tuesday. How much is left?

- 36 After spending $\frac{1}{3}$ of the money, \$200 was left. How much was there at the start?

- 37 $\frac{3}{5} + \frac{3}{4}$

from 4.03

Challenging

Questions 38–46

- 38 After spending $\frac{1}{4}$ of the money, \$60 was left. How much was there at the start?

- 39 How many $\frac{2}{3}$ m lengths can be cut from a 12 m pipe?

- 40 $\sqrt[3]{8}$

from 3.08

- 41 Jack had \$120. $\frac{1}{3}$ was spent on Monday and $\frac{1}{4}$ of what was left on Tuesday. How much is left?

- 42 $7\frac{2}{3} - 3\frac{1}{3}$

from 4.04

- 43 A garden is $\frac{1}{2}$ lawn and $\frac{1}{6}$ vegetables. The rest is flowers. What fraction is flowers?

- 44 After spending $\frac{3}{4}$ of the money, \$20 was left. How much was there at the start?

- 45 Write the reciprocal of $\frac{7}{4}$.

from 4.07

- 46 How many $\frac{3}{5}$ m lengths can be cut from a 15 m pipe?

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Use the fraction key on your calculator to enter fractions and mixed numerals. The $S \rightleftharpoons D$ key switches the answer between a fraction and a decimal. Use brackets for more than one step.

Watch your teacher.
Copy every step.

E1 Use a calculator: $\frac{5}{12} + \frac{7}{18}$

E2 Use a calculator: $2\frac{3}{8} \times 1$

$$\frac{5}{6}$$

E3 Use a calculator to write $\frac{7}{16}$ as a decimal.

Easy

Questions 1–9

1 Use a calculator: $\frac{4}{9} \times \frac{2}{13}$

2 Use a calculator to write $\frac{3}{8}$ as a decimal.

3 Use a calculator to simplify $\frac{66}{78}$.

4 Use a calculator: $\frac{7}{8} + \frac{13}{14}$

5 Use a calculator to write $\frac{3}{16}$ as a decimal.

6 Use a calculator to simplify $\frac{6}{54}$.

7 Use a calculator: $\frac{2}{5} + \frac{1}{5}$

8 Use a calculator to write $\frac{9}{25}$ as a decimal.

9 $5 \times \frac{3}{5}$

from 4.06

Easy

Questions 10–18

10 Use a calculator to write $\frac{7}{8}$ as a decimal.

11 Use a calculator to simplify $\frac{70}{150}$.

12 Use a calculator: $\frac{5}{14} \times \frac{13}{14}$

13 Round 6 708 to its first digit.

from 3.01

14 Use a calculator: $\frac{15}{16} \div \frac{4}{7}$

$$= \frac{41}{\quad}$$

15 Use a calculator to write $\frac{13}{50}$ as a decimal.

16 Use a calculator to simplify $\frac{130}{143}$.

17 How many $\frac{1}{5}$ s are in 2? Work out $2 \div \frac{1}{5}$.

from 4.07

18 Use a calculator: $\frac{18}{19} + \frac{4}{5}$

Medium

Questions 19–26

19 Use a calculator: $\frac{4}{17} \div \frac{7}{9}$

20 Use a calculator: $4\frac{4}{5} - 2\frac{2}{5}$

21 Simplify $\frac{72}{117}$ fully.

from 4.01

22 Use a calculator: $(\frac{2}{3} + \frac{4}{8}) \times \frac{5}{8}$

23 Use a calculator: $\frac{1}{7} \div \frac{9}{11}$

24 45 students: $\frac{1}{3}$ walk to school and $\frac{1}{5}$ ride. The rest take the bus. How many take the bus?

from 4.08

25 Use a calculator: $6\frac{4}{9} - 3\frac{1}{3}$

26 Use a calculator: $(\frac{1}{3} + \frac{5}{8}) \times \frac{3}{4}$

Medium

Questions 27–37

27 Use a calculator: $5\frac{1}{8} - 3\frac{7}{8}$

28 Use a calculator: $(\frac{5}{8} + \frac{1}{2}) \times \frac{8}{9}$

 29 The lines are parallel. What is the co-interior angle to 126° ?

from 2.09

30 Use a calculator: $\frac{1}{5} \div \frac{6}{19}$

31 Which is larger: $\frac{1}{12}$ or $\frac{1}{4}$?

from 4.02

32 Which is larger: $\frac{9}{13}$ or $\frac{11}{16}$? Use decimals.

33 Use a calculator: $4\frac{1}{6} - 2\frac{5}{6}$

34 $\frac{1}{5}$ of $\frac{1}{6}$

from 4.06

35 Use a calculator: $(\frac{2}{7} + \frac{1}{6}) \times \frac{2}{8}$

36 $6 \div \frac{4}{5}$

from 4.07

37 Use a calculator: $\frac{1}{6} \div \frac{3}{16}$

Challenging

Questions 38–46

38 Which is larger: $\frac{7}{9}$ or $\frac{11}{14}$? Use decimals.

39 A bag of rice holds $2\frac{1}{3}$ kg. How much rice is in 12 bags?

40 Draw an angle of 115° on the ray.

from 2.02



41 Use a calculator: $\frac{1}{9} + \frac{1}{8} + \frac{1}{72}$

42 A 14 cm ribbon is cut into 2 equal pieces. What fraction of the ribbon is each piece, and how long is it?

from 4.08

43 Which is larger: $\frac{5}{7}$ or $\frac{8}{11}$? Use decimals.

44 A bag of rice holds $3\frac{1}{3}$ kg. How much rice is in 29 bags?

45 $\frac{3}{4} - \frac{2}{5}$

from 4.03

46 Use a calculator: $\frac{1}{12} + \frac{1}{6} + \frac{1}{72}$

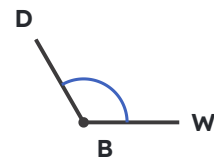
Challenging

Questions 47–56

47 A bag of rice holds $5\frac{1}{8}$ kg. How much rice is in 12 bags?

48 Use a calculator: $\frac{1}{7} + \frac{1}{8} + \frac{1}{56}$

49 Name the marked angle with three letters.



from 2.01

50 Which is larger: $\frac{13}{17}$ or $\frac{16}{21}$? Use decimals.

51 $5\frac{2}{3} - 3\frac{1}{3}$

from 4.04

52 Use a calculator: $(\frac{1}{2} + \frac{1}{2}) \times \frac{4}{6}$

53 A bag of rice holds $3\frac{1}{3}$ kg. How much rice is in 13 bags?

54 Write the reciprocal of $\frac{1}{2}$.

from 4.07

55 Use a calculator: $\frac{1}{10} + \frac{1}{6} + \frac{1}{60}$

56 Mia had \$88 and spent $\frac{1}{8}$ of it. How much is left?

from 4.08

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Per cent means out of 100. Percentage to fraction: write it over 100 and simplify. Percentage to decimal: divide by 100. Fraction or decimal to percentage: multiply by 100%.

Watch your teacher.
Copy every step.

E1 Write 35% as a fraction.

E2 Write $\frac{3}{5}$ as a percentage.

E3 Write 0.08 as a percentage.

Easy

Questions 1–9

1 Write 12% as a fraction in simplest form.

2 Write 107% as a decimal.

3 Write 0.1 as a percentage.

4 Write 60% as a fraction in simplest form.

5 Write 75% as a decimal.

6 Write 0.71 as a percentage.

7 Write 75% as a fraction in simplest form.

8 Write 5% as a decimal.

9 Write $4\frac{2}{3}$ as an improper fraction.

from 4.01

Easy

Questions 10–18

10 Write 143% as a decimal.

11 Write 0.14 as a percentage.

12 Write 5% as a fraction in simplest form.

13 Is 4 a common factor of 29 and 27?

from 3.11

14 Write $\frac{3}{5}$ as a percentage.
= $\frac{60}{100}$

= ____%

15 Write 44% as a decimal.

16 Write 0.74 as a percentage.

17 Use a calculator to write $\frac{5}{16}$ as a decimal.

from 4.09

18 Write 80% as a fraction in simplest form.

Medium

Questions 19–25

19 Write $\frac{1}{2}$ as a percentage.

20 Write 0.75 as a percentage and as a fraction.

21 Write < or >: $\frac{1}{2}$ $\square$ $\frac{3}{4}$

from 4.02

22 Order from smallest to largest: $\frac{3}{8}$, 0.4, 35%

23 Write $\frac{7}{25}$ as a percentage.

24 24 students: $\frac{1}{3}$ walk to school and $\frac{1}{6}$ ride. The rest take the bus. How many take the bus?

from 4.08

25 Write 0.04 as a percentage and as a fraction.

Medium

Questions 26–36

- 26 Write 0.9 as a percentage and as a fraction.

- 27 Order from smallest to largest: $\frac{2}{5}$, 0.45, 38%

- 28 Find the supplement of 31° .

from 2.04

- 29 Write $\frac{12}{25}$ as a percentage.

- 30 $\frac{4}{8} - \frac{1}{8}$

from 4.03

- 31 Write 12.5% as a fraction in simplest form.

- 32 Write 0.3 as a percentage and as a fraction.

- 33 How many $\frac{1}{2}$ s are in 4? Work out $4 \div \frac{1}{2}$.

from 4.07

- 34 Order from smallest to largest: 35%, 0.4, $\frac{3}{8}$

- 35 Which of these is not equal to $\frac{2}{3}$: $\frac{4}{6}$, $\frac{6}{9}$, $\frac{5}{6}$, $\frac{8}{12}$?

from 4.01

- 36 Write $\frac{4}{5}$ as a percentage.

Challenging

Questions 37–45

- 37 Write 7.5% as a fraction in simplest form.

- 38 Write $\frac{1}{9}$ as a percentage, to one decimal place.

- 39 The highest temperature was 2°C . The lowest was -6°C . What is the difference?

from 1.10

- 40 Which is the odd one out: 0.7, 70%, 7%, $\frac{7}{10}$? Explain.

- 41 Use a calculator to simplify $\frac{27}{126}$.

from 4.09

- 42 Write 2.5% as a fraction in simplest form.

- 43 Write $\frac{1}{7}$ as a percentage, to one decimal place.

- 44 $5\frac{1}{2} + 4\frac{1}{3}$

from 4.04

- 45 Which is the odd one out: $\frac{1}{2}$, $\frac{1}{5}$, 50%, 0.5? Explain.

Challenging

Questions 46–55

- 46 Write $\frac{2}{3}$ as a percentage, to one decimal place.

- 47 Which is the odd one out: $\frac{3}{5}$, 65%, 0.6, 60%? Explain.

- 48 Are the lines in train tracks parallel, perpendicular or neither?

from 2.06

- 49 Write 87.5% as a fraction in simplest form.

- 50 $\frac{4}{5}$ of 35

from 4.05

- 51 Order from smallest to largest: 38%, 0.45, $\frac{2}{5}$

- 52 Write $\frac{5}{6}$ as a percentage, to one decimal place.

- 53 $\frac{7}{9} \div 2$

from 4.07

- 54 Which is the odd one out: 0.5, 50%, $\frac{1}{5}$, $\frac{1}{2}$? Explain.

- 55 Write $\frac{25}{3}$ as a mixed numeral.

from 4.01

4.11 Percentage of a quantity

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NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Find 10% by dividing by 10, then build the percentage you need: 30% is $3 \times 10\%$, 5% is half of 10%. Or change the percentage to a fraction or a decimal and multiply.

Watch your teacher.
Copy every step.

E1 10% of 340

E2 35% of 80

E3 A \$60 jacket is 25% off.
Find the sale price.

Easy

Questions 1–12

1 10% of 570

2 What is 10% of 35?

3 50% of 87 4 10% of 680

5 What is 10% of 28?

6 50% of 17 7 1% of 600

8 What is 10% of 408?

9 Write 20% as a decimal.

from 4.10

10 50% of 25 11 50% of 4

12 What is 10% of 880?

Easy

Questions 13–22

13 What is 10% of 333?

14 100% of 71 15 50% of 94

16 $170 \div 10$

from 3.03

17 70% of 80

___ % = ___

___ % = ___

18 What is 10% of 553?

19 200% of 68

20 $\frac{3}{5}$ of 20

from 4.05

21 10% of 600

22 70% of 200

Medium

Questions 23–30

23 70% of 40

24 Lena scored 50% on a test out of 110. How many marks is that?

25 Simplify $\frac{28}{36}$ fully.

from 4.01

26 65% of \$460

27 20% of 380

28 Use a calculator: $4\frac{5}{7} - 4\frac{4}{5}$

from 4.09

29 Ruby scored 50% on a test out of 100. How many marks is that?

30 35% of \$520

4.11 Percentage of a quantity (continued)

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OF 2

Medium

Questions 31–41

- 31 Zac scored 10% on a test out of 190. How many marks is that?

- 32 65% of \$360

- 33 $\sqrt[3]{64}$

from 3.08

- 34 5% of 340

- 35 Which is larger: $\frac{1}{11}$ or $\frac{1}{6}$?

from 4.02

- 36 A \$180 jacket is 40% off. Find the sale price.

- 37 Ruby scored 20% on a test out of 70. How many marks is that?

- 38 A class has 9 girls and 16 boys. What fraction of the class are girls?

from 4.08

- 39 65% of \$240

- 40 Order from smallest to largest: 22%, $\frac{1}{5}$, 0.25

from 4.10

- 41 70% of 420

Challenging

Questions 42–51

- 42 A \$160 jacket is 15% off. Find the sale price.

- 43 Which is larger: 25% of 80 or 80% of 25?

- 44 Write DCLXXIII as a number.

from 3.06

- 45 A \$420 bike goes up in price by 25%. Find the new price.

- 46 What is half of 18?

from 4.05

- 47 A \$220 jacket is 40% off. Find the sale price.

- 48 Which is larger: 30% of 70 or 70% of 30?

- 49 $\frac{1}{3} + \frac{3}{4}$

from 4.03

- 50 A \$140 bike goes up in price by 25%. Find the new price.

- 51 Use a calculator: $(\frac{7}{9} + \frac{1}{2}) \times \frac{3}{7}$

from 4.09

Challenging

Questions 52–61

- 52 Which is larger: 20% of 45 or 45% of 20?

- 53 A \$100 bike goes up in price by 5%. Find the new price.

- 54 Evaluate 11^2 .

from 3.07

- 55 A \$110 jacket is 20% off. Find the sale price.

- 56 $6\frac{8}{9} - 5\frac{7}{9}$

from 4.04

- 57 18% of \$600

- 58 Which is larger: 40% of 15 or 15% of 40?

- 59 A 24 cm ribbon is cut into 3 equal pieces. What fraction of the ribbon is each piece, and how long is it?

from 4.08

- 60 A \$380 bike goes up in price by 25%. Find the new price.

- 61 Write $\frac{3}{4}$ as a percentage.

from 4.10

4.12 Expressing as a fraction or percentage

YEAR 7 · CHAPTER 4: FRACTIONS AND PERCENTAGES · PAGE 1 OF 2

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

Write the part over the whole as a fraction, then simplify. To make it a percentage, multiply by 100%. Change to the same units first.

E1 Write 15 out of 20 as a percentage.

E2 Write 40 cm as a fraction of 2 m.

E3 Sam got 18 out of 24. What percentage is that?

Easy

Questions 1–9

1 Write 6 out of 20 as a percentage.

2 Write 2 out of 19 as a fraction in simplest form.

3 Write 8 out of 20 as a percentage.

4 Write 1 out of 7 as a fraction in simplest form.

5 Write 4 out of 10 as a percentage.

6 Write 8 out of 20 as a fraction in simplest form.

7 Write 16 out of 50 as a percentage.

8 Write 5 out of 8 as a fraction in simplest form.

9 What is 10% of 242?

from 4.11

Easy

Questions 10–18

10 Write 14 out of 15 as a fraction in simplest form.

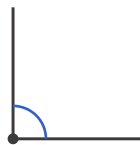
11 Write 2 out of 4 as a percentage.

12 Write 6 out of 25 as a percentage.

$$\frac{6}{25} = \frac{24}{100}$$

$$= 24\%$$

13 Classify the angle.



from 2.03

14 Write 6 out of 12 as a fraction in simplest form.

15 Write 1 out of 5 as a percentage.

16 Write 1 out of 25 as a percentage.

17 Write 49% as a decimal.

from 4.10

18 Write 3 out of 6 as a fraction in simplest form.

Medium

Questions 19–25

19 Write 3 out of 10 as a percentage.

20 Write 30 min as a fraction of 1 h.

21 Simplify $\frac{120}{156}$ fully.

from 4.01

22 Lena got 19 out of 20 on a test. What percentage is that?

23 Write 20 out of 25 as a percentage.

24 Use a calculator: $5\frac{1}{3} - 3\frac{1}{3}$

from 4.09

25 Write 10 cm as a fraction of 2 m.

4.12 Expressing as a fraction or percentage (continued)

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Medium

Questions 26–36

- 26 Write 12 min as a fraction of 1 h.

- 27 Ruby got 28 out of 40 on a test. What percentage is that?

- 28 $\sqrt[3]{8}$

from 3.08

- 29 Write 5 out of 10 as a percentage.

- 30 Which is larger: $\frac{1}{7}$ or $\frac{1}{12}$?

from 4.02

- 31 What percentage of an hour is 36 minutes?

- 32 Write 12 min as a fraction of 2 h.

- 33 200% of 69

from 4.11

- 34 Chloe got 21 out of 25 on a test. What percentage is that?

- 35 Order from smallest to largest: 0.7 , $\frac{3}{4}$, 72%

from 4.10

- 36 Write 19 out of 20 as a percentage.

Challenging

Questions 37–46

- 37 What percentage of an hour is 6 minutes?

- 38 Which is the better score: 7 out of 10 or 36 out of 50?

- 39 Find the LCM of 2, 3 and 4.

from 3.12

- 40 A class has 9 girls and 16 boys. What percentage are girls?

- 41 Use a calculator to simplify $\frac{24}{56}$.

from 4.09

- 42 What percentage of an hour is 30 minutes?

- 43 Which is the better score: 13 out of 20 or 16 out of 25?

- 44 $\frac{4}{5} - \frac{1}{3}$

from 4.03

- 45 A class has 14 girls and 11 boys. What percentage are girls?

- 46 5% of 260

from 4.11

Challenging

Questions 47–56

- 47 Which is the better score: 9 out of 10 or 22 out of 25?

- 48 A class has 9 girls and 11 boys. What percentage are girls?

- 49 For $AB \parallel CD$, what must be true of co-interior angles?

from 2.11

- 50 What percentage of an hour is 48 minutes?

- 51 $6\frac{3}{7} - 4\frac{1}{7}$

from 4.04

- 52 Zac got 21 out of 40 on a test. What percentage is that?

- 53 Which is the better score: 17 out of 20 or 21 out of 25?

- 54 Write 30% as a fraction in simplest form.

from 4.10

- 55 A class has 16 girls and 16 boys. What percentage are girls?

- 56 Use a calculator: $\frac{12}{13} \div \frac{5}{12}$

from 4.09